1. 도구 선정 근거

a. 파악한 **Test Framework**

i. Google Test (gtest)

1. 설명: Google의 C++ 테스트 프레임워크, C 코드에도 적용 가능
2. 특징 :
   a. 풍부한 테스트 기능 제공 (ASSERT, EXPECT 등 다양한 매크로 지원)
   b. 자동화된 테스트
   c. 멀티플랫폼 지원 (Linux, macOS, Windows 등)
   d. Mocking 지원 (gMock)
3. 단점 : 순수 C 코드 테스트 시 약간의 래퍼 코드가 필요할 수 있음.

ii. Check

1. 설명: C 전용 단위 테스트 프레임워크로, 간단하고 사용하기 쉬움
2. 특징:
   a. POSIX 기반 환경에서 주로 사용
   b. 테스트를 독립된 프로세스로 실행하여 충돌 시에도 다른 테스트에 영향 없음
   c. 테스트 스위트 및 테스트 케이스의 계층적 관리 가능
3. 단점: 고급 기능이 부족할 수 있음.

iii. Unity

1. 설명: 임베디드 시스템용으로 설계된 경량 단위 테스트 프레임워크
2. 특징:
   a. 메모리 사용량이 적어 리소스가 제한된 환경에 적합
   b. 단순한 구조로 빠르게 테스트 작성 가능
   c. Mocking 프레임워크(CMock)와 함께 사용 가능
3. 단점: 대규모 시스템 테스트에는 적합하지 않을 수 있음.

iv. CUnit

1. 설명: C 언어용으로 설계된 또 다른 경량 프레임워크.
2. 특징:
   a. 테스트 스위트, 테스트 케이스, 실행 결과 리포트 제공
   b. GUI와 CLI 실행 환경 지원
   c. 간단한 API
3. 단점: 업데이트가 자주 이루어지지 않아 최신 요구 사항에 부합하지 않을 수 있음

b. Unity 선정 근거

i.RVC 프로그램을 개발하기 위함이기 때문에, 임베디드 시스템용으로 설계된 테스트 프레임워크를 사용하는게 용이

ii.github에서 Clone 한 후 바로 사용할 수 있는 간단한 구조로 빠르게 테스트 환경을 구축할 수 있음

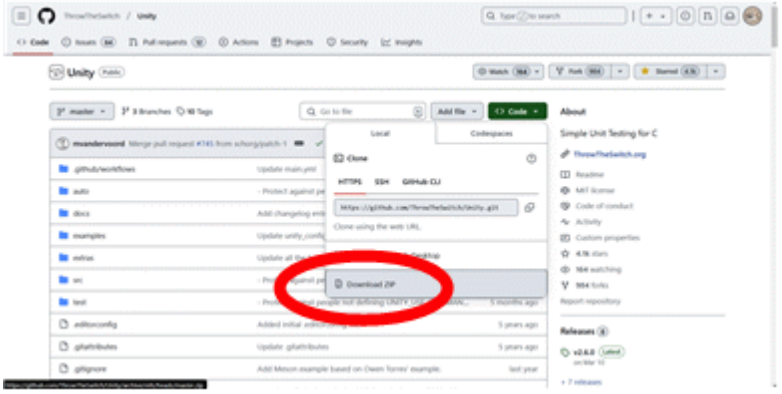
iii.소규모 프로젝트는 경량 프레임워크(Unity, Cutest) 추천

iv.많은 예제와 샘플 코드가 제공되며, GitHub 등의 저장소에서 쉽게 확인 가능 따라서 학습하는데도 용이

c. Unity 설치/실행 방법

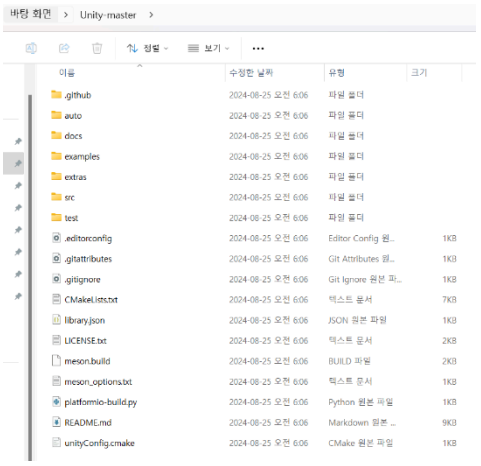
i.설치

1. Unity Github Repository : https://github.com/ThrowTheSwitch/Unity



ii.Repository를 Clone 하거나, ZIP을 다운받아도 무방

iii.Unity Test Framework 폴더 구조




```
project/
├── src/                # 프로젝트 소스 코드
│   ├── my_module.c     # 테스트할 코드
│   ├── my_module.h     # 테스트할 코드의 헤더
│   └── ...             # 다른 소스 파일들
├── test/               # 테스트 관련 파일
│   ├── unity/          # Unity 프레임워크
│   │   ├── unity.h
│   │   ├── unity.c
│   │   ├── unity_internals.h
│   │   └── ...
│   ├── mocks/          # (선택) Mock 파일 (CMock 사용 시)
│   │   ├── mock_my_module.c
│   │   ├── mock_my_module.h
│   │   └── ...
│   ├── test_my_module.c # 테스트 케이스 파일
│   ├── test_runner.c    # 테스트 실행 메인 함수
│   └── ...
├── build/              # 빌드 디렉토리 (빌드 출력물)
│   ├── test_my_module.out
│   └── ...
├── Makefile            # (선택) 빌드 스크립트
└── README.md           # 프로젝트 설명
```

1. 구조 :
   a. src/ 실제 프로젝트 소스 코드가 위치하는 디렉토리.
   b. test/모든 테스트 관련 파일이 포함된 디렉토리.
   c. unity/ 폴더: Unity 프레임워크의 소스와 헤더 파일이 포함된 디렉토리

2. unity 사용 과정 :

a. test_rvc.c

```c
#include "unity.h"
#include "rvc.h"

#include <string.h>

void setUp(void)
{
    F = L = R = D = false;
    turnCount = 0;
    isFrontMotor = false;
    cleanerMode = 0;
}

void tearDown(void)
{        You, 어제 • 1204
    // 정리 작업 필요 시 추가
}
```

setUp 함수로 각 테스트 실행 전 초기 값 설정 가능
tearDown 함수로 테스트 이후 필요한 작업 처리 가능

```c
// 정상 입력 테스트
void test_RVC_T_00(void)
{
    printf("---------------------------------\n");
    char input[] = "0000";
    input[strlen(input)] = '\0';
    outputSet result = Main_Logic(input);
    int comparison[2] = {2,2}; // 처음 실행 시 Stimulus 2 출력 + other test case
    TEST_ASSERT_EQUAL_INT_ARRAY_MESSAGE(result.outputArray, comparison, result.outputArrayCnt, "Array elements do not match!"); // 입력 유효성 확인
}
```

위 이미지와 같이 testcase 함수 작성

테스팅 결과는 rvc.c의 Main_Logic의 return값을 받아 테스트케이스의 output이 적혀있는 comparison과 비교한다.

메인 시스템의 output과, 테스트 케이스의 비교는 unity test framework에서 제공하는 TEST_ASSERT_EQUAL_INT_ARRAY_MESSAGE()를 통해 비교한다.

```c
#ifdef UNIT_TEST
int main(void)
{
    UNITY_BEGIN();
    RUN_TEST(test_RVC_T_00);

    return UNITY_END();
}
#endif
```

main 함수를 하나로 유지하기 위해 rvc.c의 기능을 Main_Logic으로 변경하고, Test을 위한 컴파일 일때는 test_rvc.c에서 main이 실행되도록 구현했다.

UNITY_BEGIN();을 통해 unity test framework을 활성화 하고,
RUN_TEST(Testcase 함수);을 통해 테스트케이스를 실행할 수 있다.
테스트가 끝나면 UNITY_END로 unity test framework를 종료한다.

3. 프로그램 실행방법

a. <u>일반 실행 컴파일</u>

: gcc -Iunity -Isrc -o test_runner src/rvc.c test/test_rvc.c unity/unity.c

b. <u>c Unit Testing 컴파일</u>

: gcc -DUNIT_TEST -Iunity -Isrc -o test_runner src/rvc.c test/test_rvc.c unity/unity.c

c. 실행

: terminal 환경에서 컴파일을 통해 생성된 test_runner.exe 경로로 이동해 ./test_runner

i.일반 실행

```
PS D:\GitClone\Unity> gcc -Iunity -Isrc -o test_runner src/rvc.c test/test_rvc.c unity/unity.c
PS D:\GitClone\Unity> ./test_runner
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0
```

ii.c Unit Testing 실행

```
OK
PS D:\GitClone\Unity> gcc -DUNIT_TEST -Iunity -Isrc -o test_runner src/rvc.c test/test_rvc.c unity/unity.c
PS D:\GitClone\Unity> .\test_runner
```

```
Enter the FLRD values as a 4-digit binary number (e.g., 1010):000a

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:466:test_RVC_T_43:PASS

-----------------------
44 Tests 0 Failures 0 Ignored
OK
```

44개의 모든 테스트 케이스에 대해 통과한 이미지

2. 방법론 선정 근거

a. 후보군

i. Black Box Testing

1. 설명: 프로그램 내부 구조를 고려하지 않고, SRS를 기준으로 테스트
2. 특징 :
   a. 사용자 중심의 테스트 가능
   b. 코드 변경에 민감하지 않음
3. 단점 : 내부 구조의 결함 탐지에 불리

ii. White Box Testing

1. 설명: 소스 코드를 기반으로 Code Coverage를 극대화하는 데 초점을 둠
2. 특징:
   a. Code Coverage를 높이는 데 유리
   b. 내부 구조의 결함 탐지에 유리
3. 단점: 코드 변경시 테스트 환경을 수정해야 함

iii. Gray Box Testing

1. 설명: Black Box Testing과 White Box Testing을 혼합한 테스트
2. 특징:
   a. SRS와 소스 코드를 모두 참고하여 테스트를 진행
3. 단점: 두 테스트 방법론의 장점을 동시에 충족하기 어려움

b. Black Box Testing 선정 근거

i. 상용 임베디드 프로그램을 개발하는 작업이므로 이용자의 요구사항을 적극적으로 반영하여야 함.

ii. 소규모 프로젝트이므로 테스트 환경에서 Code Coverage의 중요성이 상대적으로 낮음

iii. 기존에 작성된 SRS가 있으므로, Black Box Testing에서 Test case를 만들기 쉬움

Test cases 수행 결과

0 ~ 3

```
PS D:\GitClone\Unity> gcc -DUNIT_TEST -Iunity -Isrc -o test_runner src/rvc.c test/test_rvc.c unity/unity.c
PS D:\GitClone\Unity> ./test_runner
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:423:test_RVC_T_00:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0001

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON
test/test_rvc.c:424:test_RVC_T_01:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:425:test_RVC_T_02:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0011

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON
test/test_rvc.c:426:test_RVC_T_03:PASS
-----------------------------------
```

4 ~ 7

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0100

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:427:test_RVC_T_04:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0101

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON
test/test_rvc.c:428:test_RVC_T_05:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0110

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:429:test_RVC_T_06:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0111

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON
test/test_rvc.c:430:test_RVC_T_07:PASS
```

8 ~ 12

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1110

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward
test/test_rvc.c:431:test_RVC_T_08:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward
test/test_rvc.c:432:test_RVC_T_09:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1100

Cleaner OFF
RVC is Turning Right
RVC Turned Right
test/test_rvc.c:433:test_RVC_T_10:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1101

Cleaner OFF
RVC is Turning Right
RVC Turned Right
test/test_rvc.c:434:test_RVC_T_11:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:435:test_RVC_T_12:PASS
```

13 ~ 16

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1011

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:436:test_RVC_T_13:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1000

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:437:test_RVC_T_14:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1001

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:438:test_RVC_T_15:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0001

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:439:test_RVC_T_16:PASS
```

17 ~ 19

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1000

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:440:test_RVC_T_17:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1110

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward
test/test_rvc.c:441:test_RVC_T_18:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1100

Cleaner OFF
RVC is Turning Right
RVC Turned Right
test/test_rvc.c:442:test_RVC_T_19:PASS
```

20~22

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:443:test_RVC_T_20:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1000

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:444:test_RVC_T_21:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0100

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1100

Cleaner OFF
RVC is Turning Right
RVC Turned Right
test/test_rvc.c:445:test_RVC_T_22:PASS
```

23~24

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:446:test_RVC_T_23:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1000

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:447:test_RVC_T_24:PASS
```

25~26

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0000

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1000

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left
test/test_rvc.c:448:test_RVC_T_25:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:449:test_RVC_T_26:PASS
```

```
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0011

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1100

Cleaner OFF
RVC is Turning Right
RVC Turned Right

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0100

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:450:test_RVC_T_27:PASS
```

27

```
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0100

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1100

Cleaner OFF
RVC is Turning Right
RVC Turned Right

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0100

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1101

Cleaner OFF
RVC is Turning Right
RVC Turned Right
test/test_rvc.c:451:test_RVC_T_28:PASS
```

28

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0011

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:452:test_RVC_T_29:PASS
```

29

```
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0011

RVC is Moving Forward
Dust Detected , Cleaner Power UP ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0010

RVC is Moving Forward
Cleaner ON
test/test_rvc.c:453:test_RVC_T_30:PASS
```

30

31

```
----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward
```

```
Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward
```

```
Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1111

Cleaner OFF
RVC is Turning Backward
RVC Turned Backward
test/test_rvc.c:454:test_RVC_T_31:PASS
```

32 - 36

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):0

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:455:test_RVC_T_32:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):00

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:456:test_RVC_T_33:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):000

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:457:test_RVC_T_34:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1234

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:458:test_RVC_T_35:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1029

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:459:test_RVC_T_36:PASS
```

37-40

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1010

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:460:test_RVC_T_37:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):1000

Cleaner OFF
RVC is Turning Left
RVC Turned Left

Enter the FLRD values as a 4-digit binary number (e.g., 1010):00

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:461:test_RVC_T_38:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):Nega

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:462:test_RVC_T_39:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):01 1

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:463:test_RVC_T_40:PASS
```

```
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):/

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:464:test_RVC_T_41:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):*/

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:465:test_RVC_T_42:PASS
-----------------------------------
RVC is Moving Forward
Cleaner ON

Enter the FLRD values as a 4-digit binary number (e.g., 1010):000a

Error: The input must be a 4-digit binary number consisting of only 0 and 1. Please try again.
Each position corresponds to FLRD.
test/test_rvc.c:466:test_RVC_T_43:PASS

-----------------------
44 Tests 0 Failures 0 Ignored
OK
```

c. Test cases 상세 설명

RVC 00~15

4자리 binary 데이터가 입력된 환경에서 로봇청소기의 동작이 예상되는 출력과 일치하는지 확인하는 test case

RVC 16~23

8자리 binary 데이터가 입력된 환경에서 로봇청소기의 동작이 예상되는 출력과 일치하는지 확인하는 test case

RVC 24,26

12자리 binary 데이터가 입력된 환경에서 로봇청소기의 동작이 예상되는 출력과 일치하는지 확인하는 test case

RVC 25, 28

16자리 binary 데이터가 입력된 환경에서 로봇청소기의 동작이 예상되는 출력과 일치하는지 확인하는 test case

RVC 27,29,30

20자리 binary 데이터가 입력된 환경에서 로봇청소기의 동작이 예상되는 출력과 일치하는지 확인하는 test case

RVC 31

전방에 장애물이 감지되는 입력이 20번 반복되었을때, 로봇청소기가 20번 연속으로 회전 동작을 수행하고, 이후 정상적으로 Terminate 하는지 확인하는 test case

RVC 32~34

4자리 미만의 유효하지 않은 데이터가 입력되었을 때 적절히 에러 메시지를 출력하고 시스템을 종료하는지 확인하는 test case

RVC 35, 43

'0', '1' 외의 유효하지 않은 데이터가 입력되었을 때 적절히 에러 메시지를 출력하고 시스템을 종료하는지 확인하는 test case

RCV 36

입력값의 자리수가 4의 배수가 아니면서, '0', '1' 외의 유효하지 않은 데이터가 입력되었을 때 적절히 에러 메시지를 출력하고 시스템을 종료하는지 확인하는 test case

RVC 37~38

4자리 binary 데이터와 4자리 미만의 유효하지 않은 데이터가 혼입되었을 때 적절히 에러 메시지를 출력하고 시스템을 종료하는지 확인하는 test case

RCV 39, 41~43

'0', '1' 외의 유효하지 않은 string 데이터가 입력되었을 때 적절히 에러 메시지를 출력하고 시스템을 종료하는지 확인하는 test case

RVC 40

4자리 binary 데이터가 비정상적으로 분리되어 입력되었을 때 적절히 에러 메시지를 출력하고 시스템을 종료하는지 확인하는 test case

d. Homework # 3에서 수행한 테스트와의 차이점

Homework #3에서는 파일을 실행하고 직접 값을 입력해 매 입력에 대해 하나하나 테스트를 진행했다.

본 과제에서는 unity를 통해 반복적으로 테스트할 수 있는 환경을 구축하여, 메인 소스코드를 수정했을 때 이전보다 더 정확하고 빠르게 수정한 코드가 정상적으로 작동하는지 확인할 수 있다.
또한 테스트 환경을 구축함에 따라 다양한 입력에 대해 일괄적으로 빠르게 테스트를 진행할 수 있게 되었다.